

FILTER DESIGN TEST CASE

Lattice Filter - n78 (3.3-3.8 GHz 5G NR) for Public Safety Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P2 - High

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Lattice Filter filter targeting n78 (3.3-3.8 GHz 5G NR) for Public Safety Band Filter. The filter is designed on Quartz substrate with a target center frequency of 3093.5 MHz and bandwidth of 285.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Public Safety Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Lattice Filter
- Frequency Band: n78 (3.3-3.8 GHz 5G NR)
- Application: Public Safety Band Filter
- Substrate: Quartz
- Package: Flip-Chip BGA
- Design Tool: CST Studio Suite 2024
- Design Center: Singapore RF Lab

This specification applies to all variants of the Lattice Filter design targeting n78 (3.3-3.8 GHz 5G NR). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3093.5 MHz
Bandwidth (3 dB):	285.1 MHz
Insertion Loss Target:	<= 3.3 dB
Return Loss Target:	>= 8.0 dB
Out-of-Band Rejection:	>= 38.0 dB
Isolation (Duplexer):	>= 24.0 dB
Q Factor:	3943
Temperature Coefficient:	-9.5 ppm/°C
Die Size:	1.3 x 1.6 mm
Wafer Size:	6-inch
Number of Resonators:	12
Electrode Material:	Mo

Passivation: Si3N4

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the Lattice Filter template project.
2. Define the substrate stack: Quartz with specified layer thicknesses.
3. Set design targets: center freq 3093.5 MHz, BW 285.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, Quartz).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 9280 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.3 dB
- Return loss in passband shall be >= 8.0 dB
- Out-of-band rejection at +/- 428 MHz offset >= 38.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 3.7 MHz
- Power handling shall be >= +30 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 85%

6. TEST RESULTS

Measurement Date: 2025-06-02
Devices Tested: 76
Insertion Loss (meas): 2.97 dB
Return Loss (meas): 10.5 dB
Rejection (meas): 41.5 dB
Isolation (meas): 29.5 dB
Q Factor (meas): 3943
Group Delay Variation: 4.5 ns
Power Handling: +36 dBm
Die Yield: 96.7%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The Lattice Filter design demonstrated below-target performance for n78 (3.3-3.8 GHz 5G NR).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Simulation-to-measurement correlation was within 3% for all key parameters.
- e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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